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Parameter Optimization for Deriving Bluetooth-based Social Network Graphs

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Abstract—Pervasive technologies such as Bluetooth (BT) are capable of detecting close proximity. As a result, they are increasingly used for deriving social networks. However, the validity and reliability of the inferred networks is questionable as evaluation procedures are often omitted. In this paper, we consider the process of deriving and evaluating a Bluetooth derived network as a parameter optimization problem. Using the BNEA algorithm, we investigate the effect of the *number of detected connections*, *time window* in which these are detected, and the *direction* of the resulting connection. Our results confirm the importance of conducting a throughout evaluation procedure when deriving social networks based on BT data. Going through the parameter optimization process, we are able to increase the accuracy of the derived BT networks by a maximum of 10%, compared to deriving the networks without it. Our outcomes indicate that reducing the false positives can be achieved by setting a particular *connection_weight*. Furthermore, with the *window_size* parameter we show that more BT observations does not necessarily mean more accurate networks. With respect to the *connection_type*, we observe the accuracy of deriving undirected networks is higher than the accuracy of deriving directed networks. Finally, based on the outcomes we are able to come up with a set of recommendations for the future developers of similar BT data collection systems.

Index Terms—pervasive computing, bluetooth, social networks, performance evaluation, algorithms

I. INTRODUCTION

A large number of empirical studies have been conducted in the field of IoT, on collecting rich and multimodal data for interpreting human behaviour. Being part of personal electronic devices like smartphones or laptops, and integrated in many IoT devices, Bluetooth (BT) is one of the most prominent pervasive technologies for acquiring proximity traces. BT data is usually considered as ‘natural’ way of collecting human social networks. This intuition comes from the fact that BT is present in the most frequently used day-to-day personal technological devices.

However, inferring social networks from BT data comes with significant challenges. First, the collected data is rather

noisy and may contain substantial measurement error [23]–[26]. When neglected, these errors are propagated to the next phase of analysing the properties of the inferred networks. Another apparent drawback is the uncertainty about the ground truth (GT) data when inferring social networks. As we show in Section II, the majority of previous empirical studies ignore the GT when inferring social networks. They assume that the collected data *are* the network itself, while in fact, the inferred networks are derived based on the measured interactions, and are strongly dependent on the quality of the collected experimental data. Finally, many network inference methods are relying on parameter thresholds that are hand-tuned or based on domain expert knowledge. As a consequence, the presented inference processes are suitable only for ad-hoc, specific network scenarios. There is an evident lack of studies offering more generic methods or guidance for inferring social networks based on BT-based (or other pervasive) systems. Furthermore, the threshold selection process (being based on trial-and-error fashion) produces many negative results, that are unfortunately rarely reported by the previous studies. By presenting only the final description of the network and its parameters, the readers lose valuable information on the whole process of inferring the social networks, that can be beneficial for the design of future BT-based data collection systems.

In this paper, we address the above mentioned challenges, and consider the process of inferring BT-based social networks as a parameter optimization problem. We present a Bluetooth Network Evaluation Algorithm (BNEA) for optimal inference of social networks from noisy BT data. The algorithm delivers the best estimate of the underlying network based on a brute-force search space of a pre-defined set of input parameters. The final parameter threshold selection is derived by evaluating all the inferred BT networks with GT networks, rather than hand-tuning or relying on expert knowledge. All the possible inferred BT networks are labeled with particular *accuracy* as a success criteria measure coming from the evaluation process.

The three BNEA input parameters are defined considering a set of reliability challenges with regard to the BT

technology itself. The *connection_weight* parameter tackles the issue of false positives in the network-inferring process [17], as BT-derived proximity can be an error-prone indicator of social connection [27]. The aim of this parameter is to find the optimal number of pairwise BT connections (or optimal weight) that represent a real-world social connection, rather than encounter by chance. The *window_size* parameter explores the BT data's temporal dimension, with objective to determine the right duration of collecting BT data for making the best estimate of the real network structure. The effect of representing BT connections as directed or undirected edges in the social network graph is explored with the *connection_type* parameter. This parameter gives an idea of the influence of a particular network design choice on the network representation's accuracy.

We test our approach on a BT dataset of pairwise relationships in school classes. The BNEA algorithm's goal is to infer optimal (accurate) social networks representing friendship ties among the participants. The BT observations are obtained from in-the-wild experiments from participants' smartphones, unlike the majority of previous studies that are based on data from devices that are not commercialized nor used in real-life scenarios.

The main contributions of this paper are:

- We present a methodology for inferring and evaluating BT-based social networks via the BNEA algorithm
- We demonstrate a throughout analysis of the network inferring process, based on a real-life smartphones data
- We provide a set of guidances aimed towards the future developers of similar BT data collection systems, based on the limitations of the BT technology for representing social connections

The remainder of the paper is organized as follows. In the next section we discuss previous research on BT-based pervasive systems, and show the limitations of their experimental setup and the BT technology itself. In Section III we introduce the dataset used in this work. The methodology behind deriving both the GT and BT networks is explained in Section IV. In Section V we present the BNEA algorithm and the algorithm's input parameter sets. The results based on the algorithm's outputs are thoroughly presented in Section VI. Finally, we discuss the influence of the parameter optimization process on the derived BT networks' accuracy in Section VII and conclude the presented work.

II. BACKGROUND

This section provides an overview of previous studies that have relied on BT for understanding social connections. We provide evidence that the validation of the used systems is most often omitted. Moreover, we point out the limitations of the BT technology and emphasize the necessity of understanding these constraints when relying on BT-based pervasive systems for research experiments.

Any methodology for data collection by human carried devices, suffers from bias caused by the technology involved in the process and the particular use that users make of the

device. Understanding the limitations of the BT technology for characterizing human behavior is a first step towards developing reliable data collection system, and was already investigated by [16], [17]. The main identified problems are around person-device uncertainty, the granularity of the BT traces and the particularities of the BT technology itself. The person-device uncertainty problem relates to the fact that the proximity traces detected by devices do not necessarily correspond to what is happening in real life. It is common to find false positive BT logs in situations when people forget to carry their device or are coincidentally in proximity with others. The granularity of the traces is directly related to the quality of the data analysis. In addition, smartphones come with different configurations and operating systems and can have a specific set of restrictions when it comes to using the BT technology. This makes the collection of proximity traces data more challenging. Moreover, long-term continuous BT scans can drastically reduce the devices' battery, and increase the risk of experiment dropout.

These limitations are commonly overlooked when performing analysis on the noisy BT datasets. The scientific conclusions based on the inferred networks are made without considering the BT technological constraints. These studies have demonstrated the versatility of BT-based collected data for recognizing social patterns, inferring social relationships, creating networking structures [1]–[6] or even influencing mental health [7]. In addition, BT data has been leveraged for context-oriented opportunistic networking applications and epidemic modeling [28], [29], [31]. The outcomes of the social networks inference have been applied to many domains for providing improved networking services [2], designing communication overhead algorithms [4], or modeling social distance measures [28]. However, there is an universal lack of precise evaluation methods in order to confirm the meaningfulness of the inferred social networks, *before* performing analysis or deriving scientific conclusions based on them.

We identify and present several approaches for inferring networks from data, based on previous research. These are found over different application domains and rely on particular knowledge to infer and measure the quality of the inferred networks. Different validation strategies for calculating the accuracy of the inferred networks have been proposed. Our interest is a specific subset of network inference, where nodes are known and the task is to perform edge inference. Edge inference is commonly interpreted as evaluating models via prediction; or inferring parameters on an assumed parametric model on the data. The goal of predictive methods [8], [35]–[37] is to model some predictive aspects of the underlying data in order to infer relationships between entities. On the other hand, the parametric models usually rely on a domain knowledge base in order to construct the networks. These models make assumptions on the edge distribution, for example by leveraging maximum likelihood estimation [31]. Another category are non-parametric models that use statistical tests to determine edge weights [32]–[34].

III. DATASET

The data has been collected as part of the MyMovez project, by means of a research app on a dedicated smartphone at 21 schools [18]. The participants had the MyMovez app installed on dedicated smartphones, that they were using besides their (possible) private devices during the experiment.

The data was gathered in five waves, each consisting of five consecutive days, out of which three school days and two weekend days (not in this particular order), labeled as D1-D5. For this experiment we rely on the data from the three initial waves: February/March 2016 (Wave 1 - W1), April/May 2016 (Wave 2 - W2), June/July 2016 (Wave 3 - W3).

A. BT-Networks dataset

The BT data collection framework has been programmatically implemented as part of the MyMovez app. The app scans and detects other participants' phones that are present in the BT range of approximately 10 meters. During each day, a total 50 BT scans for nearby devices are performed. The scans run every quarter-hour, starting at 07:00 until 19:00. We label the scan periods as S0-S49. The time span of each scan period varies between 3-5 minutes, during which nearby devices are able to detect each other. Each BT entry (row) in the database contains the following information: School, Class, Wave, Day, Date, Time, Child_ID, Detected_Child_ID.

We have performed a data cleansing procedure in two steps. In the first step, we have dismissed the classes in which less than 60% of pupils have participated in the experiment, resulting in 26 out of 196 initial classes. The subsequent analysis showed us that the BT data collection process varied significantly among the 26 classes: the number of collected BT entries per class ranges between 358 and 19229. Therefore, a subset of classes have been selected where the data collection is reasonable, and upon which valid conclusions can be made. The class with ID 67 is the principal point of the analysis described in the remainder of the paper. In order to test the generalizability of the BNEA algorithm, we have additionally included four other classes with IDs: 71, 72 (same school with class 67) and 130, 136 (belonging to another school).

Table I gives more details about the selected classes. In addition to what is shown, we observed that the data is not evenly distributed during the 5 days of data collection per wave (valid for all classes). Understandably, weekend days generated much less (or no) data, compared to school days. Moreover, not all 50 defined scan periods resulted in BT data being collected. Again, this was observed at all the selected classes. An example is shown in Fig. 1, depicting the scan periods distribution of *Class67*. BT observations were obtained at only 29 out of 50 possible scan periods per day. The remaining 21 scan periods had no BT data collected. We have concluded that the success of a scan period was overlapping with particular events happening during that time of the day.

For the *Class67* participants, school day begins at 08:20 in the morning (represented by S5 and S6, at 08:00 and 08:15), first break time is from 10:00 to 10:25 (corresponding with S13 and S14, starting at 10:00 and 10:15). The second break

TABLE I
CLASSES DATASET DESCRIPTION.

ClassID	Participation (%)	Bluetooth Observations (#)			
		Total	W1	W2	W3
67	18/29; 62%	19229	5560	9558	4111
71	20/30; 67%	17468	11117	3050	3319
72	20/29; 69%	12145	10347	1484	314
130	20/25; 80%	3734	1794	1822	118
136	19/28; 68%	3165	1822	755	588

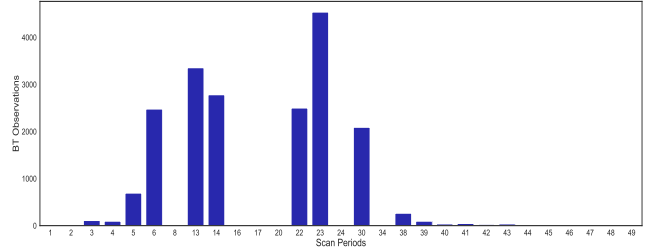


Fig. 1. The successful scan periods of *Class67* (aggregated over all waves).

starts at 12:05 until 12:35 (S22 and S23), while the school day ends at 16:05 as represented by S38 in Fig. 1. The scan periods during the lectures time were rarely resulting in BT data being collected. A possible explanation is that participants were not actively using their smartphones, resulting with the app process being stacked in the background or killed by the smartphone's OS.

B. GT-networks dataset

A set of nomination questions on friendship and social connections was used in order to derive the GT networks. The participants were asked to complete a set of 16 peer nomination (sociometric) questions at the beginning of each wave. For example, participants were asked to nominate peer(s) from their class that they ask for advice, they consider as leader, they are friends with, they respect, they hang out with, and they want to be like. The exact set of nomination questions can be found in [18].

IV. BUILDING THE SOCIAL NETWORKS

A. Bluetooth-Based Social Network

The BT networks are inferred upon the BT data observations described in Section III. We define the connection's weight between the two nodes i and j as:

$$w_{i,j} = \frac{\text{num_connections}(i,j) + \text{num_connections}(j,i)}{\text{num_scans}(i) + \text{num_scans}(j)}. \quad (1)$$

where $\text{num_connections}(i,j)$ is the number of scan periods in which node i detected node j , and $\text{num_scans}(i)$ is the total number of scan periods that resulted in at least one successful scan of node i . Note that $w_{i,j} = w_{j,i}$ for each connection.

However, calculating the weight by using this equation does not compensate for the difference in total number of scans,

among the node pairs. Fig. 2's scan table demonstrates the essence of this problem - the significant difference in the total number of scans generated by each participant's phone, during the defined period (in this case W2). The labeled squares indicate the assortment of connections obtained between participants. Calculating the weight as based on (1), results with same value ($w=1$) for pair 983, 973 as pair 965, 973, even though the $num_scans(983) + num_scans(973) = 44$ while $num_scans(965) + num_scans(973) = 5$. In order to give certain level of confidentiality of the detected connection, and normalize the weight coefficients according to the level of scan participation, we have implemented Laplace smoothing on these values [19]. Laplace smoothing is a mathematical technique that can be used in order to smooth categorical data. Using this method we introduce two parameters to our weight calculation, α and β that represent our estimation of what weight is appropriate if we have no information about the BT observations. The α and β parameter values should be fitted to the particular problem scenario. In our case, we have selected $\alpha = 0$ and $\beta = 1$ as this setting biases towards 0, meaning the weight value is reduced, even more with smaller num_scans values. This results in an improved formula which is founded on a statistical estimator:

$$w_{i,j} = \frac{num_connections(i,j) + num_connections(j,i) + \alpha}{num_scans(i) + num_scans(j) + \beta}. \quad (2)$$

The weights are once again in range between [0,1], however in this case we obtain well-balanced weights based on the scanning frequency of the nodes pair.

Intuitively, the BT networks can be considered as undirected. Device I locates device J, gives information about their mutual discovery in the physical space, no matter if for some technical reason device J can not detect device I. In the undirected BT networks, there is a single edge between two nodes. In the directed BT network representation (as required for the *edge_type* parameter) there exist an edge from i to j with $w_{i,j}$, and another edge in the opposite direction with strength $w_{j,i}$, with same weights.

B. Ground Truth Social Network

Each participant can nominate multiple persons on any of the 16 nomination items. The label $q_n nomination_{i,j}$ represents the nomination of node i to node j on the particular question q_n .

First, we describe the modeling of the directed GT network. Given two nodes i and j , the edge $e_{i,j}$ directed from i to j , has weight $w_{i,j}$ based on the number of nominations that node i gave to node j . Having an edge $e_{i,j}$, does not mean that edge $e_{j,i}$ exists as well. For example, an influential individual might not nominate all the nodes that have nominated him/her. The weight of the directed edge from node i to node j is given by (3).

$$w_{i,j} = \frac{\sum_{n=1}^{16} q_n nomination_{i,j}}{16}. \quad (3)$$

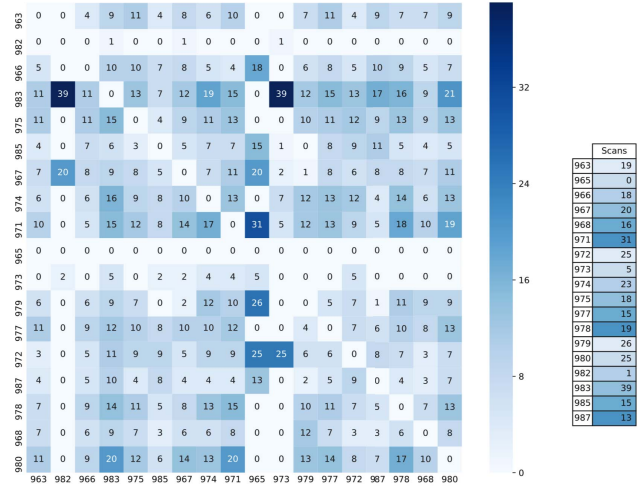


Fig. 2. An example connection heatmap between all participants of the class 67. This heatmap is based on the W2 dataset. The participants are randomly labeled with three digits starting with 9. Each square of i row and j column is labeled with $num_connections(i,j)$. The table on the left gives information on the number of successful scan periods (num_scans) per participant.

where $\sum_{n=1}^{16} q_n nomination_{i,j}$ is the sum of all nominations that node i gave to node j .

In the second network representation we consider the GT network as undirected. In this case the mutual nominations between node i and j are summed up and divided by 32 to obtain the strength of their relation. In both network scenarios the GT network is weighted in range [0,1].

V. BLUETOOTH NETWORK EVALUATION ALGORITHM

We have developed an algorithm for inferring and evaluating the BT networks, the BNEA. The goal of the algorithm is to find the best solution (the optimal set of input parameters) for deriving the optimal BT network. We consider three (input) parameters for optimization:

- 1) *connection_weight* identifies the set of genuine edges in the derived BT networks, rather than random BT encounters. Only the edges with $w > connection_weight$ are included in a particular inferred BT network.
- 2) *window_size* parameter slices a subset of BT observations for inferring a BT network. The window size can be altered on day and scan period(s) during a particular day. This parameter estimates the number of BT observations needed in order to have an good representation of GT network.
- 3) *connection_type* – we conceptualize both directed and undirected connection types in the social networks. This parameter depicts the consequence of a particular design approach on the accurate representation of the inferred social networks from data.

The BNEA algorithm was executed on the input set of parameters defined in Table II. The value of *connection_weight* was restricted to maximum of 0.6. Setting higher values of *connection_weight* resulted in unrealistic outcomes, with

TABLE II
INPUT PARAMETERS FOR THE BNEA ALGORITHM

Parameter	Values
<i>connection_weight</i>	[0.0 - 0.6] with stepsize 0.01
<i>window_size(day)</i>	D1, D2, D3, D4, D5
<i>window_size(scan_period)</i>	S3, S4, S5, S6, S13, S14, S22, S23 S30, S38, S39, S40, S41, S42, S43
<i>connection_type</i>	directed, undirected

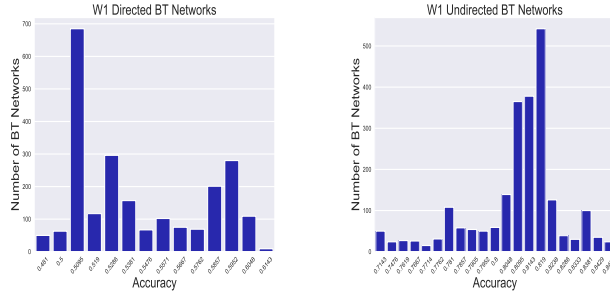


Fig. 3. Distinct accuracies and the corresponding number of BT networks, obtained from the BNEA algorithm (for *Class67*). Both W1' directed and undirected BT networks and their accuracy are displayed.

many edges being dismissed and as a consequence creation of isolated nodes (nodes without edge connectivity) in the BT networks.

The algorithm generates *connection_weight* values with step size of 0.01, resulting in a input set of 60 values. The *window_size* parameter was defined on both day and scan period granularity. All five days were taken into consideration by using the combination formula $\binom{n}{k} = {}^nC_k = \frac{n!}{k!(n-k)!}$, with $n = 5$ and $k = 1$, resulting in the following combinations: (D1), (D1 and D2), (D1 and D2 and D3) etc. In addition, the day granularity was enriched with the scan periods that appeared during a particular combination of day(s). As the BT data is collected unevenly among the scan periods, only the non-trivial scans (at least 30 BT observations per scan) were considered. As a result, we have used 16 scan periods with labels as shown in Table II. The scan periods were combined in a similar manner like days: (S3), (S3 and S4), (S3, S4, ... SN). The *connection_type* parameter had two variations: undirected and directed networks. The BNEA algorithm was run separately for each of the three data waves (W1, W2 and W3). Each wave represents a distinctive data collection process executed at different time period of the year. A single GT network is generated for each wave and *connection_type*. On contrary, a large set of BT networks, based on the input parameter ranges defined in Table II are derived for each wave and *connection_type*.

In the second phase of the BNEA algorithm, each inferred BT network is evaluated with the corresponding GT network, for each wave and *connection_type*. For example, the set of directed BT networks obtained for W1 are evaluated on the single directed GT network build on W1 data as well.

We use *accuracy* as the BNEA algorithm's evaluation measure. Accuracy is a good metric when evaluating binary classification problems, with (nearly) balanced classes, as our GT networks are. We have chosen accuracy over other evaluation metrics such as precision or recall, as in the context of our problem we have no preference towards either minimizing the false positives or false negatives. The accuracy is calculated as follows:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where *TP* (True Positive) represents the number of edges found in the BT network, also existing in the GT network; *TN* (True Negative) represents the number of edges not found in the BT, also not existing in the GT network; *FP* (False Positive) represents the number of edges found in the BT network, but not existing at the GT network; and *FN* (False Negative) represents the number of edges not found in the BT network, but are existing at the GT network. The accuracy value varies in the range of 0 to 1.

VI. RESULTS

The BNEA algorithm was run at five school classes, as already defined in Section III. The presented analysis focuses on *Class67*, but we also investigate the generalizability of the algorithm by including four additional classes. The algorithm was run separately per class, data wave and *connection_type*. First, we present the magnitude of obtained accuracies based on the input parameter set. There is a high level of isomorphism at the inferred BT networks, as expected given the granularity of the BNEA input parameter set. Fig. 3 shows one example of the distribution of the inferred BT-networks for *Class67* based on W1. Total 14 different BT networks (accuracy values) are inferred as directed BT networks (depicted on the x-axis). Many BT-networks have the same accuracy (are isomorphic), for example, 685 derived BT networks have an accuracy of 0.5095 and 9 BT networks have the maximum accuracy of 0.6143.

Table III gives a more detailed overview of the obtained accuracy ranges for all the five classes. In addition to the accuracy range, we give information on the number (#) of distinct accuracy values (or distinct BT network representations) obtained for each wave and *connection_type*.

The obtained accuracy ranges follow a similar pattern among the different classes and waves. This is the first implication that the algorithm is generalizable among different data sets (classes). The results show that there can be many possible network representations of friendship connections derived from BT data. The number of possible inferred networks vary from 4 to 70 among different classes, data waves and *connection_type*. Selecting the 'right' BT network is therefore challenging, as wrong design decisions can lead to selection of the non-optimal BT network. For example, choosing a particular directed BT network for *Class67* at W1 gives us 15 selection options. Particular decision can lead to a

TABLE III
ACCURACY RANGE OF THE INFERRED BT NETWORKS

Class	W1		W2		W3	
	DIR	UND	DIR	UND	DIR	UND
67	[0.4809, 0.6142] #15	[0.6047, 0.7238] #23	[0.4901, 0.5882] #16	[0.5130, 0.6503] #41	[0.4889, 0.6066] #17	[0.6323, 0.7423] #32
71	[0.3782, 0.5962] #16	[0.7052, 0.8026] #38	[0.5196, 0.6437] #20	[0.7450, 0.7843] #13	[0.4736, 0.5789] #19	[0.6754, 0.7923] #38
72	[0.4707, 0.6169] #25	[0.5555, 0.7573] #70	[0.5392, 0.5980] #10	[0.6666, 0.7091] #13	[0.4181, 0.4909] # 5	[0.6000, 0.6545] # 7
130	[0.4807, 0.6089] #11	[0.6410, 0.7243] #14	[0.4444, 0.5029] #11	[0.5789, 0.6315] #19	[0.5777, 0.6444] # 4	[0.7333, 0.7666] # 4
136	[0.3782, 0.5961] #16	[0.6730, 0.8397] #24	[0.3333, 0.4743] #12	[0.5961, 0.7051] #18	[0.5000, 0.6636] #10	[0.6909, 0.8545] #19

TABLE IV
OPTIMAL PARAMETER SETS ACHIEVING HIGHEST EVALUATION ACCURACY PER CLASS

Class	Type	W1			W2			W3		
		Parameters	Acc	#BT	Parameters	Acc	#BT	Parameters	Acc	#BT
67	DIR	W:0.40;D:2;S:12	0.6143	5050 (91%)	W:0.54;D:2;S:4	0.5882	4550 (48%)	W:0.19;D:2;S:8	0.6066	1972 (48%)
67	UND	W:0.40;D:2;S:9	0.7238	4110 (74%)	W:0.20;D:5;S:3	0.6503	713 (7%)	W:0.42;D:2;S:7	0.7463	1375 (33%)
71	DIR	W:0.11;D:5;S:9	0.5962	1734 (16%)	W:0.54;D:5;S:8	0.6438	2032 (67%)	W:0.18;D:2;S:7	0.5789	1409 (42%)
71	UND	W:0.05;D:2;S:10	0.8026	9170 (82%)	W:0.35;D:5;S:8	0.7843	2032 (67%)	W:0.11;D:2;S:0	0.7924	2897 (87%)
72	DIR	W:0.19;D:2;S:0	0.6170	8319 (80%)	W:0.24;D:5;S:11	0.5980	1432 (96%)	W:0.33;D:3;S:9	0.4909	157 (50%)
72	UND	W:0.16;D:2;S:10	0.7573	7115 (69%)	W:0.37;D:2;S:9	0.7092	615 (41%)	W:0.45;D:3;S:9	0.6545	157 (50%)
130	DIR	W:0.39;D:5;S:7	0.6090	1412 (79%)	W:0.21;D:2;S:9	0.5029	1783 (98%)	W:0.59;D:3;S:8	0.6444	93 (79%)
130	UND	W:0.42;D:5;S:8	0.7244	1773 (99%)	W:0.47;D:2;S:10	0.6316	1795 (99%)	W:0.59;D:3;S:8	0.7667	93 (79%)
136	DIR	W:0.11;D:5;S:8	0.5962	1734 (95%)	W:0.12;D:2;S:10	0.4744	713 (94%)	W:0.24;D:3;S:8	0.6636	499 (85%)
136	UND	W:0.07;D:5;S:0	0.8397	1882 (100%)	W:0.12;D:2;S:10	0.7051	713 (94%)	W:0.24;D:3;S:8	0.8545	499 (85%)

difference of accuracy of up to 13%, in case the BT network with minimum accuracy is selected.

Evidently, our results show that undirected BT networks are more alike their GT networks counterpart, compared to the directed BT networks. The ranges of obtained accuracies at undirected BT networks varies from approx 51% - 85%. On contrary, the directed BT networks have accuracies of 33% to 66%.

More than one BT network might be labeled with maximum accuracy, as we already saw in Fig.3. The question that arises is how do we select the single optimal BT network in this case, and the corresponding optimal parameter set? In case the optimal accuracy is equivalent among two or more BT networks, we turn to the number of BT observations that are used to derive those BT networks. The BT network that was generated with lesser BT entries is selected as optimal. In case of another tie at the number of BT observations, we rely on the number of days and scan periods required to derive the networks. In this final case, the optimal BT network is the one being created with lesser number of days and/or scan periods.

Next we show the BNEA's optimal parameter value sets that achieved the highest accuracy when deriving the BT networks. These are the parameters that are used in order to infer the BT networks most similar to the corresponding GT networks. Table IV reports on the optimal parameters for each class, wave and *connection_type*. The table's *Parameters* column shows the optimal parameter set in the form (W, D, S) representing the *connection_weight* value, followed by *window_size*'s number of days and number of scan periods, respectively. For example, in order to derive the optimal W1 directed BT network of *Class67* (with *accuracy* = 0.6143),

we should use the BT observations from first two days and 12 scan periods, and only include BT network's edges with weights bigger than 0.40. As already reported, the achieved accuracies are considerably higher (around 20%-25%) at undirected BT networks. The highest optimal accuracy is obtained in the case of *Class136* W1 undirected BT network with *accuracy* = 0.8545. The social connections derived in this BT network represent approximately 85% of the connections in the corresponding *Class136* W1's undirected GT network. On contrary, the lowest optimal accuracy is obtained at *Class72* W2 directed BT network with *accuracy* of 0.4909.

The optimal *connection_weight* value varies among waves and network types. We should emphasize that the optimal *connection_weight* parameter is always higher than 0.0. This confirms the applicability of using *connection_weight* when inferring BT networks. Having a certain *connection_weight* value removes part of the BT data noise. The optimal *window_size* in order to infer the optimal BT networks is often consisting in less than 5 (the max number of days in our dataset). This implies that bigger *window_size*, i.e. more days of data collection doesn't necessarily mean more accurate BT networks. We observe that two days of collecting BT data is enough to infer 50% of the optimal BT networks (with 15/30 times), followed by five days with 30% of cases (9 times) and three days with 20% of cases (6 times).

We also analyse the number of BT observations (#BT) that are used in order to obtain the most accurate BT networks. As there is a significant variation of the BT data collection process among the classes (see Table I), we present the percentages of used BT observations per class to derive the optimal BT networks, rather than reporting on the absolute number of BT

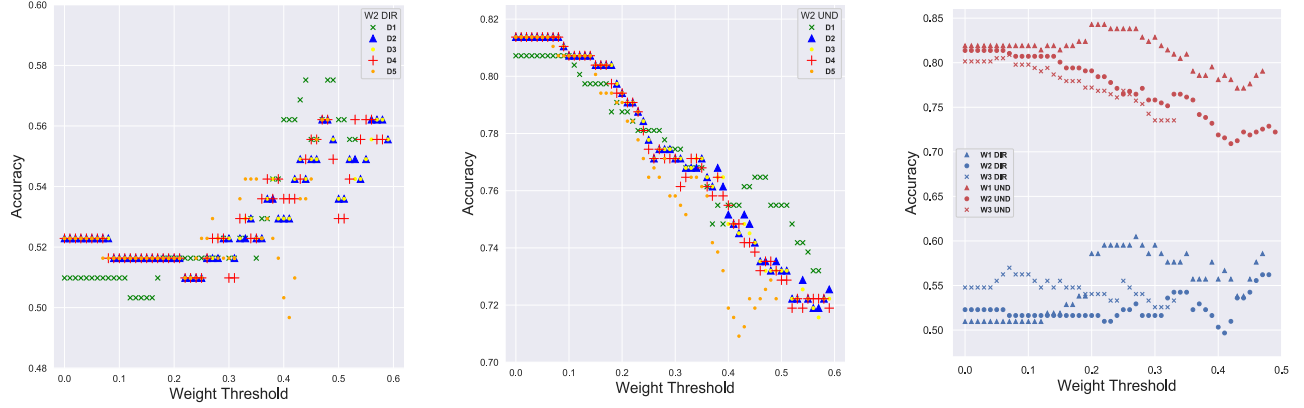


Fig. 4. BNEA algorithm's parameter optimization search space in three distinct scenarios of *Class67*

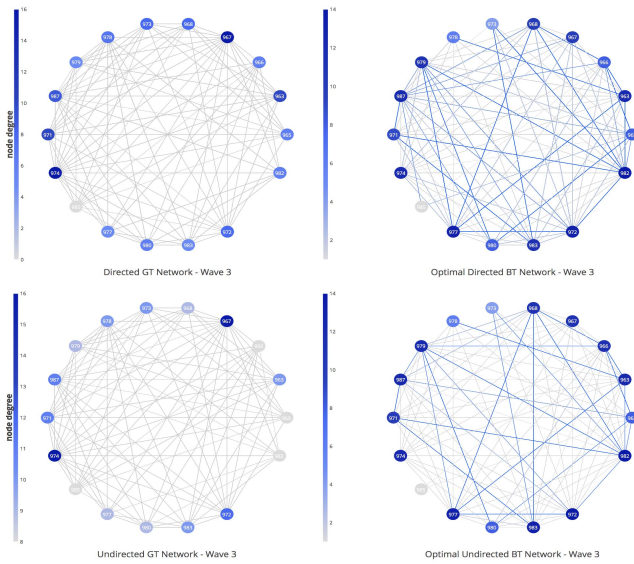


Fig. 5. Structural network difference between *Class67*' W3 GT and optimal BT networks. The upper image shows the comparison in case of directed networks, followed by the undirected networks comparison below.

observations. These percentages are present in Table IV next to the number of BT observations for a particular case. *Class130* and *Class136* need at least 79% of the collected BT data to infer the optimal networks. This is expected as the data collection at these two classes is significantly lower compared to *Class67*, *Class71* and *Class72*. There is a bigger variety in the results of the latter classes, for which the data collection process was more successful. Rarely, more than 80% of the collected BT data was needed (at only 5 of 18 cases). Most commonly we observe that about 40-70% of the BT data was capitalized (at 9 cases). In two exceptional scenarios only 7% and 16% of the BT data was enough for inferring the optimal BT networks.

We finalize this section by focusing on the *Class67* outcomes. First, several runs of the BNEA algorithm are visually represented in order to get a better understanding of the algorithm's execution. Next, the densities of the optimal BT networks obtained for *Class67* are compared to their GT counterparts. This shows that the BNEA algorithm has a certain overestimation in the number of inferred edges. Lastly, the no-optimized networks are compared with the optimal BT networks, showing the improvement of accuracy the BNEA provides.

The graphical representation of three particular scenarios of running the BNEA with *Class67* is displayed in Fig. 4. In the first scenario (left image) the BNEA is run for deriving the *Class67*' directed BT networks of W2. The central image (the second scenario) shows the results of running the BNEA algorithm for inferring the W2 undirected BT networks. We examine the differences in BT networks' accuracies as we increase the *window_size*'s day parameter. The days' combinations exhibit similar accuracy patterns, however from both figures it is noticeable that having five days of BT data (D5) rarely outperforms the other day's combinations when fewer days are included. This is confirmed at the optimal parameter sets discussed in the paragraph above. In contrary, one day of data (D1) is more commonly the best local optimum solution. Another noteworthy pattern is the reverse nature of obtained accuracies at directed and undirected BT networks. At derived directed BT networks the accuracy increases as the *connection_weight* increases, while the opposite stands for the undirected BT networks. The left most figure removes the *window_size*'s day granularity (here the total five days are considered) and displays the obtained accuracies for all waves and *connection_type* combinations. Even though the data is collected at different waves (different time periods of the year), we identify visible similarities in the obtained BT network accuracies regardless of the wave. This is particularly visible by comparing the obtained accuracies at W2 and W3 at both directed and undirected BT networks.

TABLE V
NETWORK DENSITY COMPARISON

Type	W1 Density		W2 Density		W3 Density	
	GT	BT	GT	BT	GT	BT
dir	0.50	0.64	0.46	0.46	0.48	0.55
dir	0.50	0.65	0.46	0.38	0.48	0.60
dir	0.50	0.67	0.46	0.39	0.48	0.35
und	0.69	0.76	0.67	0.79	0.64	0.67
und	0.69	0.77	0.67	0.76	0.64	0.63
und	0.69	0.74	0.67	0.75	0.64	0.62

Table V gives a comparison between the densities of the GT networks and the *Class67* BT networks derived from the optimal parameters defined in Table IV. The densities of the corresponding BT and GT networks are comparable. In most cases, the BT networks have more edges compared to the GT networks, as the BT network's densities are usually slightly higher. We visualize an example of the density differences in Fig. 5. The figure depicts W3's GT and optimal derived BT networks, in both directed and undirected *edge_type*. The blue lines in the BT networks (on the right side) illustrate these supplementary edges (social connections) that were derived during the BNEA execution. The nodes are colored by their node degree represented by the number of connections a pupil has in the classroom. The node degrees have bigger variety at the GT networks, but on the other hand the nodes are more interconnected in the BT networks, therefore the higher density.

Finally, we compare the accuracies at the above described optimal scenarios (OS) with the baseline scenarios (BS) where no parameter optimization is applied. In the BS, the BT networks are derived without removing any edges (*weight_threshold* = 0) and all the BT observations are taken into consideration (*window_size* : *day* = 5, all scan periods included). Therefore, in this scenario the BNEA algorithm is not relevant. Compared to the BS, the optimal parameter input set improves the accuracy of the derived BT networks at all data waves. At the W1 directed network the BS achieves an accuracy of 0.5095 compared to 0.6142 for the OS, in the case of undirected networks the difference is 0.8190 for BS and 0.8476 for OS. The W2's directed BS has accuracy of 0.5228, as opposed to 0.5882 for OS. The W2 undirected BT network is the only case in which the accuracy is not improved, both scenarios have an equivalent accuracy of 0.8137. Finally, the W3 directed BS derives accuracy of 0.5477, as opposite to 0.6066 for OS, the undirected BT networks with minimal improvement (BS: 0.8014, OS: 0.8161).

VII. DISCUSSION AND CONCLUSIONS

In this paper we have presented a methodology for evaluating social networks inferred from BT data and performed a detailed analysis of the network inferring process. Ultimately, the described analysis can serve as a guidance for developers of similar BT-based data collection systems. Our parameter selection was driven by the limitations of the BT technology itself.

The *connection_weight* was used in order to question the significance of the discovered BT connections as representatives of a real-life social connection. We argued that some of the naively derived BT connections might be a result of a random encounter. All of the optimal parameter sets derived from the BNEA algorithm, in Table IV, imply that removing a certain percentages of edges improves the accuracy of the network. Therefore, setting a particular *connection_weight* threshold lowers the number of false positives in the networks. Another evident outcome is the large diversity of optimal *connection_weight* values. This means that the parameter has to be calculated on a per-case bases, as the level of noisiness of the data is different per scenario.

Using the *window_size* parameter we learned that having more BT observations does not necessarily result in more accurate BT networks. On the contrary, we showed that two days of data collection is frequently a sufficient period to derive BT networks with optimal accuracy. Proper time granularity of the data collection process can be essential for both researchers and participants of an experiment. For the researchers, shorter data collection time makes the system less error prone and eases the data analysis process. For the participants, it makes the whole experience more pleasant, as smartphone battery is saved, they spend less time in participating in the experiment, and therefore the risk of dropout is reduced.

Finally, we looked at the effect of a particular *connection_type* on the accuracy of the inferred networks. Undirected BT networks are much closer to their GT counterparts, being approximately 20% more accurate compared to the directed BT networks. However, we argue that these outcomes might be biased by the nature of the social networks, as undirected networks have higher density, therefore reducing the chance for detecting false positives. Therefore, it is important for researchers to consider the potential biases based on their design decisions, and aim for selecting the right network representation for the problem that is being solved.

The BNEA algorithm delivers a set of possible optimal BT inferred networks for each scenario (wave and *connection_type*). Evidently, the inferring process results in many possibilities of representing a social network. So finding the 'right' network needs to be based on some criteria. In our case, we define optimal inferred networks as those who are achieving the highest accuracy and are obtained from lesser BT observations, and/or lesser days of data collection. The accuracies of the optimal BT networks also vary (between 48% - 85%). We conclude that the accuracy is dependent on the quality of the data collection (days, BT observations) and the network connection type.

This study also has some limitations. First, we have worked with significant volume of missing data. This is an evident issue especially relevant for real-world experiments that tend to be error-prone and no flawless data collection can be expected. Finding new ways of dealing with missing data by using machine learning or applying an alternative pervasive technology (in addition to Bluetooth) should be considered. Furthermore, even though for this research we consider the

traditional survey methods as ground truth, they can be erring as well. Human-made mistakes in answering questionnaires can result in erroneous social network graphs. Moreover, the notions of social influence can be larger than only considering friend-based nominations, as we do in this research. For example by social norms: a person can already be influenced by other people, that he/she did not nominate, but are in near proximity. An inevitable limitation comes from the nature of the BT technology with the drawbacks explained earlier in this paper. Using the rich IoT ecosystem with the most recent pervasive technologies like wearables or BLE based systems [22] can additionally improve the reliability of inferring network representations from data.

To conclude, the obtained results emphasize the need for making the right design decisions before deriving BT-based social network graphs. Being able to use reliable pervasive-based technologies like BT for deriving real-life social networks can reduce the overhead of the traditional data collection methods like questionnaire or surveys. Humans are surrounded by technology, therefore technology-based data collection can seem more natural, compared to the traditional ways of gathering data. It is important to realize that pervasive systems will always result in an *approximation* of real-life social networks, similarly to questionnaire-based social network graphs. More evaluative studies on the reliability of using similar pervasive technologies can contribute to a better understanding of the proposed systems, and a further replacement of the traditional methods for deriving social networks.

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